

## Henrique Laureano

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Data Scientist | AI Engineer | Statistician | M.Sc. in Statistics and Numerical Methods in Engineering

### Professional Summary

**Staff Data Scientist and Statistician** with a decade of experience bridging advanced mathematical theory and enterprise execution. Specialist in architecting agentic AI architectures, real-time fraud engines, and multi-level probabilistic forecasting for global leaders (EBANX, Volvo). Graduate-trained in Bayesian modeling and optimization, translating high-dimensional financial data into automated decision frameworks and multi-million-dollar bottom-line impact.

### Experience

*EBANX – Curitiba-PR, Brazil*

#### **Staff Data Scientist & AI Engineer (Finance & Operations)**

[May 2025 - Present]

**Enterprise Agentic AI:** Architect and deploy Zenith, an agentic AI system serving C-Level, FP&A, Treasury, Value Creation, and Country Growth teams to automate P&L drivers, gross profit analysis, and granular PCVM (Price, Cost, Volume, Mix) diagnosis via Slack.

*Stack:* GCP (Vertex AI, Vector Search, BigQuery, Spanner, Cloud Run, Cloud Functions, Firestore), Google ADK, Python, REST/RPC APIs.

**Probabilistic Forecasting:** Design multi-level short- and long-term daily transaction volume models across markets, product lines, merchants, and payment methods using Bayesian inference and uncertainty quantification for probabilistic time-series forecasting.

**Real-Time Fraud & Risk Systems:** Engineer and maintain high-throughput, low-latency ML infrastructure for real-time fraud detection. Build production feature engineering pipelines, PySpark batch workflows, and Gradient Boosting models integrated into microservice and Feature Store architectures.

*Stack:* AWS (SageMaker, EMR, Lambda, Step Functions, DynamoDB, EC2), FastAPI, Redis/Valkey, API Gateway, CloudWatch.

#### **Senior Data Scientist (Credit Risk), Banco Bradesco – Curitiba-PR, Brazil**

[Mar 2025 - Apr 2025]

Development, validation, and monitoring of credit risk models.

*Stack:* **Languages:** Python, R, SAS, PySpark. **Cloud/Infrastructure:** On-premises (Hive, Hadoop), Databricks, MLFlow. **AI/ML:** PyTorch. **DevOps:** GitHub Actions, Bitbucket.

*Volvo Trucks Latin America – Curitiba-PR, Brazil*

#### **Lead Data Scientist**

[Jun 2023 - Feb 2025]

Led architecture and end-to-end delivery of advanced AI initiatives across LATAM. Concurrently acted as Data & AI Product Manager for one quarter, driving strategic project prioritization and executive roadmap alignment.

#### **Senior Data Scientist (Advanced Analytics)**

[Sep 2022 - May 2023]

Delivered major financial impact for LATAM HQ, generating seven-digit (9%) annual savings via improved S&A expense forecasting and six-digit monthly reductions in service costs through high-frequency telemetry

modeling. Additionally, deployed Bayesian GAMMs for volume forecasting (+10% accuracy, <5% error) and optimized service contract pricing, driving a 5% sales increase and a 10% reduction in churn.

*Stack:* **Languages:** Python, R, SQL, PySpark. **Cloud/Infrastructure:** On-premises (SQL, Hadoop), Databricks, MLFlow. **AI/ML:** PyTorch, Stan, INLA. **DevOps:** GitHub Actions.

**Senior Statistician**, *Pelé Pequeno Príncipe Research Institute* – Curitiba-PR, Brazil

[May 2021 – Sep 2022]

Largest exclusively pediatric hospital in Latin America. Led data visualization, sampling design, and predictive modeling for complex, infectious, and oncogenic diseases. Managed statistical workflows for epidemiological and clinical studies, as well as bioinformatics applications.

*Stack:* **Languages:** Python, R, SQL. **Cloud/Infrastructure:** On-premises (SQL), Linux. **AI/ML:** PyTorch, JAX, Stan, INLA. **DevOps:** GitHub Actions.

**Statistical Consultant**, *Freelance Statistical Consulting*

[Jan 2014 – Sep 2022]

End-to-end statistical support for over 20 master's and PhD theses.

- **Diverse Field Expertise:** Agronomy, Biology, Economics, Nursing, Electrical and Chemical Engineering, Pharmacy, Education, Psychology, Chemistry, Medicine, and Zootechnics.
- **Core Focus:** Translating complex research questions into robust probabilistic frameworks, including data cleaning, hypothesis testing, and advanced modeling in **R and Python**.

**Tech Stack:** **Languages:** Python, R. **AI/ML:** PyTorch, Stan, INLA. **DevOps:** GitHub Actions.

**Statistician**, *Telehealth Center – Hospital das Clínicas (UFMG)* – Belo Horizonte-MG, Brazil

[Jul 2016 – Dec 2016]

Analyzed echocardiographic data in tropical diseases, developing predictive and statistical models to quantify risk factors for Chagas and rheumatic diseases. Special focus on competing risks and multi-state survival models. Left to pursue an MSc in Statistics at UNICAMP.

## Education

**Masters in Statistics and Numerical Methods in Engineering**, *Universidade Federal do Paraná (UFPR)*

[2021]

**Interrupted PhD in Applied Mathematics and Computational Sciences (AMCS)**,  
*King Abdullah University of Science and Technology (KAUST)*

[2019]

**Interrupted Masters in Statistics**, *Universidade Estadual de Campinas (UNICAMP)*

[2017]

**Bachelor in Statistics**, *Universidade Federal do Paraná (UFPR)*

[2016]

## Publications & Research

- **Peer-Reviewed Research:** Published **16 scientific papers** in international journals covering diverse fields such as Statistics, Medicine, and Engineering.
- **Google Scholar Profile:** [Full Publication List & Citations](#).

## Licenses & Certifications

- **Certified SAFe® Product Owner / Product Manager**, Issued by Scaled Agile, Inc. (2023).
- **EF SET English Certificate (C1 Advanced)**, Issued by EF Education First. (2026).